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Semester Project

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GLOBAL CLIMATE CHANGE DATA WAREHOUSE & BI DASHBOARD

Project Report — Business Intelligence & Data Warehousing

Power BI | PostgreSQL | Python ETL | Star Schema

Platform	Microsoft Power BI Desktop + PostgreSQL
Data Source	public.fact_climate_global
ETL Language	Python 3.x (pandas, psycopg2, numpy)
Schema Type	Star Schema (3 dimensions + 1 fact table)
Dashboard Pages	3 pages, 11 visualizations+ 1 filter
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1. Problem Statement

Problem Statement

Climate change data is abundant but fragmented, technically complex, and largely inaccessible to decision-makers without specialized tools. Organizations and researchers struggle to derive actionable insights from raw climate records because:

- Climate indicators (temperature, CO_2, sea level, extreme events) are stored in flat tables without analytical structure, making multi-dimensional analysis difficult.
- No interactive dashboards exist to explore trends by country, time period, or energy profile — forcing analysts to rely on slow, static spreadsheet reports.
- Key performance indicators (KPIs) such as climate risk index, temperature anomaly, and extreme event scores are not pre-computed, requiring manual calculation.
- There is no geospatial layer to visually identify the most climate-vulnerable nations at a glance, hindering prioritization of international support and policy response.

This project addresses these challenges by designing and implementing a full data warehousing and BI pipeline: from raw PostgreSQL data through an ETL process into a star-schema warehouse, culminating in an interactive Power BI dashboard that makes climate insights accessible to all stakeholders.

2. Business Requirements & Use Case

Primary Use Case

Environmental researchers, policy analysts, and academic institutions need a self-service BI tool to monitor global climate trends, compare country-level environmental performance, and identify high-risk regions — without requiring SQL or programming knowledge.

Business Requirements

- BR-01: The system must support interactive filtering by country and year with instant visual refresh.
- BR-02: Users must be able to compare CO_2 emissions, temperature trends, and energy profiles across countries in a single view.
- BR-03: The dashboard must display at least one geospatial visualization showing geographic climate risk distribution.
- BR-04: All KPI metrics must be pre-computed and available as aggregated measures in the BI layer.
- BR-05: The ETL pipeline must be re-runnable and idempotent — repeated runs must not create duplicate records.
- BR-06: The warehouse must follow a star schema to enable OLAP-style slicing and dicing across time, country, and energy dimensions.

OLAP Operations Supported

OLAP Operation	Description	Example
Roll-Up	Aggregate to higher level	Year → Decade → Era
Drill-Down	Go to finer detail	Region → Country → Year
Slice	Filter on one dimension	Year = 2020
Dice	Filter on multiple dims	Year 2010-2020, Country = Pakistan
Pivot	Cross-tabulate	Country × Decade → Avg Temp

3. Dataset Description

Source

The dataset originates from a PostgreSQL database table: **public.fact_climate_global**. It contains one record per country per year, capturing 14 environmental and climate-risk measurements.

Fact Table — public.fact_climate_global

Field	Type	Description
year	Integer	Year of measurement
country	Text	Country name
global_avg_temperature	Decimal	Mean surface temperature (°C)
co2_concentration_ppm	Decimal	CO2 in parts per million
heatwave_days	Integer	Annual heatwave days
flood_events_count	Integer	Number of flood events
drought_index	Decimal	Drought severity (0–1 scale)
sea_level_rise_mm	Decimal	Sea level rise (mm)
renewable_energy_share	Decimal	Renewable % of total energy
fossil_fuel_consumption	Decimal	Total fossil fuel consumed
air_quality_index	Decimal	AQI (higher = worse)
forest_cover_percent	Decimal	% land covered by forest
deforestation_rate	Decimal	Annual deforestation rate
climate_risk_index	Decimal	Composite vulnerability (0–1)

Dimension Tables (Star Schema)

- **dim_time**: time_id, year, decade, era, is_current_decade
- **dim_country**: country_id, country, region, is_developing
- **dim_energy**: energy_id, energy_type, description, is_clear

4. Objectives

KPIs

- Average Global Temperature per decade
- CO_2 Concentration trend (ppm) by year
- Climate Risk Index by country (geo-mapped)
- Renewable Energy Share vs. Fossil Fuel Consumption
- Extreme Event Score (weighted: heatwaves + floods + drought)
- High-Risk Country Count (climate_risk_index > 0.7)

OLAP Goals

- Enable roll-up from country level to regional and global aggregates
- Support drill-down from decade to year for trend analysis
- Allow slicing by year range via interactive slicers in Power BI
- Enable cross-country comparison on all environmental metrics

Dashboard Goals

- Page 1 — Climate Indicators: Temperature, CO_2, heatwaves, floods, drought
- Page 2 — Energy & Environment: Sea level, renewable energy, air quality, deforestation
- Page 3 — Geographic Risk Map: Azure Maps choropleth of climate risk index

5. Methodology

ETL Steps

Stage	Action	Key Tools / Output
1. Extract	Read from PostgreSQL source	tablepsycopg2, pandas
2. Clean	Normalise, cast types, fill nulls,	cap outlierspandas, numpy
3. Transform	Build dim_time, dim_country, d	im_energy, fact_climatepandas DataFrames
4. Load	Bulk insert to warehouse sche	ma (upsert)execute_values, SQL DDL

Warehouse Schema — Star Schema

The warehouse uses a Star Schema with a central fact table (fact_climate) linked to three conformed dimension tables via surrogate integer keys. Two derived columns (temp_anomaly and extreme_event_score) are computed during the Transform stage and stored in the fact table for fast BI queries.

BI Techniques

- Line charts for time-series trend analysis (temperature, CO2, sea level, deforestation)
- Clustered bar charts for country-level comparisons (CO2, air quality)
- Treemap for proportional area visualization (drought index by country)
- Stacked area chart for cumulative trend (sea level rise)
- Donut chart for part-to-whole analysis (forest cover by country)
- Azure Maps geospatial visual for geographic risk mapping
- Slicers (year & country) for interactive cross-filtering
- DAX measures for KPI aggregation (AVERAGE, SUM, COUNTROWS, FILTER)

6. Expected Outcomes

Dashboards

A 3-page interactive Power BI dashboard accessible to non-technical stakeholders, covering climate indicators, energy and environment, and geospatial risk — with year and country slicers for dynamic exploration.

Insights

- Global average temperature shows a clear upward trend across decades, confirming warming patterns.
- Countries with higher fossil fuel consumption correlate with elevated CO2 levels and lower air quality.
- Renewable energy adoption is growing but remains below 50% for most high-emission countries.
- Sea level rise is accelerating, with the steepest increases observed post-2000.
- The climate risk index identifies specific geographic clusters (Sub-Saharan Africa, South/Southeast Asia) as most vulnerable.
- Deforestation continues at a significant rate in tropical regions, reducing natural carbon sinks.